



CSE 320 - Computer Networks

LAB Session A

09.03.2026

Subnetting and Static Routing Between Multiple Networks

A. Further Network Exercise

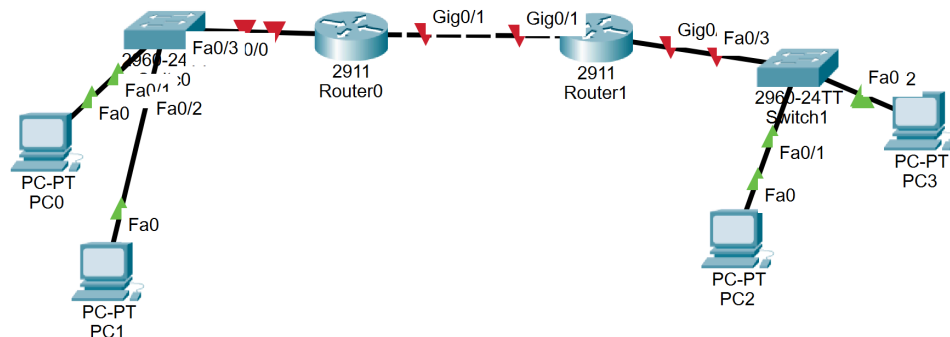
In this lab, you will learn how to:

- divide a Class C network into smaller subnetworks
- assign those subnetworks to router and switch interfaces
- configure end devices with proper IP addresses
- connect two routers using a separate transit network
- add **static routes** so different subnetworks can communicate

This lab moves from simple VLAN work into actual **Layer 3 network design**.

1. Topology

Careful about the interface plan :)



2. Addressing task

Base Network:

192.168.50.0/24

You will divide it into **four equal /26 subnets**.

Each /26 subnet has:

- **64 total addresses**
- **62 usable host addresses**

Subnet Mask:

255.255.255.192

3. Subnet names and assignment

Subnet Name	Network Address	Usable Host Range	Broadcast	Purpose
Engineering LAN	192.168.50.0/26	192.168.50.1 – 192.168.50.62	192.168.50.63	PC0 network
Student LAN	192.168.50.64/26	192.168.50.65 – 192.168.50.126	192.168.50.127	PC1 network
Server LAN	192.168.50.128/26	192.168.50.129 – 192.168.50.190	192.168.50.191	PC2 and PC3 network
Transit Network	192.168.50.192/26	192.168.50.193 – 192.168.50.254	192.168.50.255	Router-to-router link

4. Full addressing table

Router interfaces

Device	Interface	IP Address	Subnet
R1	G0/0	192.168.50.1	Engineering LAN
R1	G0/1	192.168.50.65	Student LAN

R1	G0/2	192.168.50.193	Transit Network
R2	G0/0	192.168.50.129	Server LAN
R2	G0/1	192.168.50.194	Transit Network

PC addresses

Device	IP Address	Mask	Default Gateway
PC0	192.168.50.10	255.255.255.192	192.168.50.1
PC1	192.168.50.70	255.255.255.192	192.168.50.65
PC2	192.168.50.140	255.255.255.192	192.168.50.129
PC3	192.168.50.150	255.255.255.192	192.168.50.129

5. Cabling guide

Make the following connections:

Left side

- PC0 → SW1 Fa0/1
- PC1 → SW1 Fa0/2
- SW1 Fa0/24 → R1 G0/0
- R1 G0/1 can connect directly to another small switch/PC if you want a physically separate Student LAN

Right side

- PC2 → SW2 Fa0/1
- PC3 → SW2 Fa0/2
- SW2 Fa0/24 → R2 G0/0

Router-to-router link

- R1 G0/2 → R2 G0/1

6. VLAN Design

Switch0 VLANs:

10 -> Engineering LAN, 20 -> Student LAN

Switch1 VLANs:

30 -> Server LAN

7. Configure Switch0

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Engineering
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name Students
Switch(config-vlan)#interface fastEthernet 0/1
Switch(config-vlan)#interface fastEthernet 0/2
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
Switch#
```

8. Configure Switch1

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
Switch(config-vlan)#name Servers
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fastEthernet0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fastEthernet0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
Switch#
```

9. Configure Router0 (Router-on-a-Stick)

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.50.1 255.255.255.192
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.50.65 255.255.255.192
Router(config-subif)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config)#interface g0/1
Router(config-if)#ip address 192.168.50.193 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config)#end
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

10. Configure Router1

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
interface g0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.50.129 255.255.255.192
Router(config-subif)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config)#interface g0/1
Router(config-if)#ip address 192.168.50.194 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config)#end
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

11. Configure Static Routes

Router0

```
Router0(config)#
Router0(config)#ip route 192.168.50.128 255.255.255.192 192.168.50.194
```

Router1

```
Router1(config)#
Router1(config)#ip route 192.168.50.0 255.255.255.192 192.168.50.193
Router1(config)#ip route 192.168.50.64 255.255.255.192 192.168.50.193
Router1(config)#
```

12. Verify

Check router interfaces

Check the following table with “show ip interface brief”

```
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.30	192.168.50.129	YES	manual	up	up
GigabitEthernet0/1	192.168.50.194	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Check routing table

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.50.0/24 is variably subnetted, 6 subnets, 2 masks
S       192.168.50.0/26 [1/0] via 192.168.50.193
S       192.168.50.64/26 [1/0] via 192.168.50.193
C       192.168.50.128/26 is directly connected, GigabitEthernet0/0.30
L       192.168.50.129/32 is directly connected, GigabitEthernet0/0.30
C       192.168.50.192/26 is directly connected, GigabitEthernet0/1
L       192.168.50.194/32 is directly connected, GigabitEthernet0/1

Router#
```

13. Test Connectivity with ping

Don't forget to enter the PC ip address, subnet mask, and gateway address

